

Failure-Mode Discovery and Mitigation via Contrastive Captioning on the Waterbirds Benchmark

A Closed-Loop Unsupervised Pipeline for Spurious Feature Discovery

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Institution Indian Institute of Technology Guwahati
Course Computational Cognitive Machine Learning (CCML)
Benchmark Waterbirds (WILDS) — 5,794 held-out test images
Year 2026

+12.68%	75.92%	80.84%	297
Worst-Group Gain	Best Worst-Group Acc	Group 2 Accuracy	Error Images Named

ABSTRACT

Empirical Risk Minimization (ERM) in deep vision models systematically exploits spurious correlations between target labels and background context, resulting in catastrophic failure on minority subgroups. This report presents a fully unsupervised, closed-loop pipeline that: (i) trains a base ResNet-50 classifier to expose structured failure modes on the Waterbirds benchmark; (ii) discovers coherent error clusters in CLIP embedding space via UMAP dimensionality reduction and HDBSCAN density-based clustering; (iii) assigns natural-language names to each cluster through contrastive captioning against an 86-phrase curated word bank; and (iv) mitigates worst-group failures via GroupDRO retraining on the discovered text slices — without requiring any human-annotated subgroup labels during training. Evaluated on 5,794 held-out test images, the pipeline achieves a **+12.68% absolute worst-group accuracy gain** (63.24% $\rightarrow$ 75.92%) over ERM, more than doubling the gain of artificial random-slice baselines (+6.16%), and boosts the confounded minority group (Waterbird on Land) from 63.2% to 80.84%.

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Role	Domain	Key Deliverable	Parallel Start
Role A	Model and Data	<code>predictions.csv</code> (schema: <code>image_id</code> , <code>filepath</code> , <code>true_label</code> , <code>pred_label</code> , <code>is_error</code> , <code>split</code>) — 348 validation errors	Day 1 (critical path)
Role B	Embeddings and Clustering	<code>clusters.csv</code> — 14 HDBSCAN clusters discovered from CLIP ViT-L/14 embeddings, UMAP 40-D reduction	After Role A handoff
Role C	Language and Naming	<code>slice_names.csv</code> + <code>slice_membership.csv</code> — 14 slices named via contrastive CLIP scoring over 86-phrase word bank	Day 1 (word bank unblocked)
Role D	Evaluation and Fix	GroupDRO harness — +12.68% worst-group gain achieved with 8-slice pruned set, early stopping at Epoch 50	Day 1 (harness unblocked)

Data Contract (Day 1 Agreement): All inter-role data exchange is keyed strictly on `image_id` as the sole immutable join key. Programmatic assertions are enforced at every merge to prevent row-misalignment. The `predictions.csv` row order is fixed and never re-sorted, acting as ground-truth alignment for Role B's embedding generation.

1. Introduction

Modern deep learning classifiers achieve exceptional average-case performance on standard benchmarks, yet frequently rely on *spurious correlations* — incidental statistical associations between the target label and confounding context — rather than learning genuinely invariant features. On the Waterbirds benchmark, a ResNet-50 trained with ERM learns to associate waterbirds with water backgrounds and landbirds with land backgrounds, achieving 88.70% average accuracy while catastrophically misclassifying the minority subgroup of waterbirds placed on land backgrounds at only **63.24%**.

The central challenge is that identifying and labelling failure modes at scale requires expensive human annotation of subgroup memberships. This project implements a fully automated, closed-loop pipeline that discovers, names, and mitigates spurious failure modes using only the model’s own prediction errors and pre-trained vision-language models (CLIP and BLIP) — without accessing any ground-truth subgroup labels during training. The key hypothesis is that coherent clusters in the CLIP embedding space of misclassified images correspond to semantically meaningful spurious features that can be named via contrastive captioning and then mitigated via GroupDRO retraining.

2. Related Work

Reference	Contribution	Relation to This Work
Sagawa et al. (ICLR 2020)	Introduced GroupDRO, showing overparameterised ERM fails on minority subgroups.	Provides the GroupDRO algorithm used in Phase 4; requires ground-truth group labels — our work removes this.
Radford et al. (ICML 2021)	CLIP: joint image–text embedding space via contrastive pretraining on 400M pairs.	CLIP embeddings form the basis of Phases 2 and 3 (clustering and contrastive naming).
d’Eon et al. (FAccT 2022)	Spotlight: automated slice discovery using vision-language models without human labels.	Directly motivates the pipeline; this project extends the concept with HDBSCAN and GroupDRO.
Li et al. (ICML 2022)	BLIP: bootstrapped vision-language model for image captioning and understanding.	Used in Phase 3 to mine background-descriptive phrases for the word bank from Waterbirds captions.
Koh et al. (ICML 2021)	WILDS: curated benchmark of in-the-wild distribution shifts including Waterbirds.	Provides the dataset, standardised splits, and group metadata used for evaluation-only comparison.

3. Dataset and Problem Setting

3.1 The Waterbirds Benchmark

Waterbirds is a controlled benchmark for studying spurious correlations, constructed by compositing CUB-200 bird images onto Places background scenes. It encodes a 2×2 factorial structure between the target label (waterbird vs. landbird) and the background context (water vs. land), creating four ground-truth groups with extreme train-set imbalance:

Group	Description	Train Share	Test Count	Role in Benchmark
Group 0	Landbird on Land	$\sim 73\%$	2,255	Majority (easy)
Group 1	Landbird on Water	$\sim 4\%$	2,255	Minority
Group 2	Waterbird on Land [WORST]	$\sim 4\%$	642	Worst group — spurious failure
Group 3	Waterbird on Water	$\sim 19\%$	642	Majority (easy)

The extreme class–background imbalance means an ERM-trained model learns the shortcut “water background = waterbird” rather than learning actual bird morphology, causing catastrophic failure on Group 2 — waterbirds placed on land backgrounds.

3.2 Data Ingestion: Hugging Face Migration

The pipeline was migrated from the filesystem-dependent `wilds` package to the Hugging Face `datasets` library (`grodino/waterbirds`). The key advantages are:

- **In-Memory Access:** All three splits (train, val, test) are concatenated into a single in-memory table. Images are fetched dynamically as decoded PIL objects.
- **Immutable Join Key:** The `image_id` is the sole join key across all pipeline stages. Programmatic assertions are enforced at every merge to prevent silent row-misalignment.
- **No I/O Bottleneck:** Eliminates frequent download timeouts, broken local paths, and filesystem dependency issues that plagued the original `wilds` package integration.

3.3 Problem Formulation

Formally, the dataset contains N images with label $Y \in \{0, 1\}$ (landbird / waterbird) and a confounding context variable $C \in \{0, 1\}$ (land / water background). Standard ERM minimises the average loss over all N training images, learning the spurious correlation $P(Y=1 \mid C=1)$ instead of the true invariant feature mapping. The four groups are defined as

$$g = 2Y + C,$$

and worst-group accuracy is defined as the minimum accuracy across all four groups on the held-out test set.

4. System Architecture and Team Parallelization

4.1 Pipeline Overview

The end-to-end system consists of four sequential phases, each owned by a dedicated team role and producing a well-defined handoff artifact: (1) Base Classifier (ResNet-50, Role A) → `predictions.csv` (348 errors); (2) CLIP Embeddings + UMAP + HDBSCAN Clustering (Role B) → `clusters.csv` (14 clusters); (3) Contrastive Naming via BLIP + CLIP Word Bank (Role C) → `slice_names.csv`, `slice_membership.csv`; (4) GroupDRO Retraining (Role D) → Robust Model (+12.68% worst-group gain). A feedback loop from Phase 4 back to Phase 2 enables targeted sub-slicing for Group 2.

4.2 The Parallelization Strategy

The core challenge is the inherently sequential data dependency: clustering requires embeddings, embeddings require errors, errors require a trained classifier, naming requires clusters, and GroupDRO requires named slices. A naive split would block three roles for the entire first week. The solution was to identify which work is truly independent:

- **Role C — Word Bank (Day 1 unblocked):** Building the background-descriptive phrase list requires only the Waterbirds images (for BLIP captioning), not the classifier output.
- **Role D — Harness (Day 1 unblocked):** The GroupDRO training harness and evaluation scripts can be written and tested against dummy slice assignments before real slices arrive.
- **Role A — Critical Path:** Classifier training is the only true blocker. Once `predictions.csv` is produced, Roles B and C immediately begin in parallel.

4.3 Six Major Pipeline Improvements Over Original Plan

#	Improvement	Original	Final
1	CLIP Encoder	ViT-B/16 (512-D)	ViT-L/14 LAION-2B (768-D)
2	Clustering Algorithm	K-Means + Elbow	HDBSCAN (density-based, noise-aware)
3	Dimensionality	Raw 768-D space	UMAP 40-D (cosine metric)
4	Mean-Centering	Renormalise after subtraction	Remove renormalisation (bug fix)
5	Data Source	wilds filesystem package	Hugging Face datasets (in-memory)
6	Cluster Pruning	None	Agglomerative semantic dedup (0.25 threshold)

5. Phase 1: Base Classifier Training and Error Extraction

Role A — Model and Data. The objective of Phase 1 is to establish the data pipeline, fine-tune a base classifier, and generate the `predictions.csv` handoff artifact containing all error cases for downstream processing.

5.1 Model Selection Rationale

ResNet-50 was deliberately chosen as the base architecture — not for maximum accuracy, but because its convolutional inductive bias provides a rich, structured failure surface:

- **CNN Texture Bias:** CNNs preferentially learn local texture and background statistics over global semantic features, making them susceptible to the land-vs-water background shortcut.
- **Sufficient Error Volume:** ResNet-50 produces hundreds of errors (348 on validation), providing statistically sufficient sample sizes for HDBSCAN clustering. A near-perfect ViT would produce only 10–20 errors — too few for density-based clustering.
- **Literature Alignment:** ResNet-50 is the canonical baseline in the original GroupDRO and WILDS benchmark papers, enabling direct comparison.

5.2 Architecture: Residual Learning

ResNet-50 is a 50-layer deep convolutional neural network. The key architectural feature is the residual (shortcut) connection that solves the vanishing gradient problem. Instead of forcing each block to learn a full mapping $H(x)$, the network learns the residual:

$$F(x) = H(x) - x$$

During backpropagation, identity shortcuts allow gradients to flow directly through layers without vanishing, ensuring stable fine-tuning on standard GPU hardware without requiring massive compute resources.

5.3 Training Results

The model was evaluated strictly across isolated splits to maintain dataset discipline. The fine-tuned ResNet-50 achieved the following performance:

Metric	Validation Split	Test Split
Total Samples Evaluated	4,715	2,358
Total Failure Samples (Errors)	348	173
Average Accuracy	92.62%	92.66%
Error Rate	7.38%	7.34%
Errors Available for Clustering	348 (passed to Role B)	173 (held-out)

Conclusion: The $\sim 92.6\%$ average accuracy yields hundreds of discrete failure cases. This confirms a sufficient error pool to pass to Role B for CLIP embeddings and clustering. The 348 validation errors are the primary input to Phases 2 and 3.

5.4 Handoff Artifact: predictions.csv

The inference output was compiled according to the exact schema agreed on Day 1:

```
image_id | filepath | true_label | pred_label | is_error | split
```

Row order is strictly fixed (never re-sorted), acting as the ground-truth alignment for Role B's `embeddings.npy` generation to prevent the critical row-misalignment risk identified in the Day-1 planning session. All downstream joins are performed on `image_id`, never row index.

5.5 Role A — Key Design Decisions

Decision	Rationale
Use validation split errors for clustering	Validation split is larger (4,715 vs 2,358) giving 348 errors vs 173. More errors = better clustering density for HDBSCAN.
Include both error and non-error rows in CSV	Role D needs the full assignment for GroupDRO training. Role B filters to <code>is_error==1</code> internally.
ImageNet pretraining, not random init	Transfer learning converges faster on Waterbirds; random init would require far more epochs to produce meaningful spurious correlations to cluster.
Fine-tune all layers (not frozen backbone)	Full fine-tuning allows the model to adapt its feature representations to the specific background cues in Waterbirds.

6. Phase 2: Embedding, Dimensionality Reduction and Clustering

Role B — Embeddings and Clustering. Phase 2 transforms the 348 error images from `predictions.csv` into semantically grouped clusters, producing `clusters.csv` as the handoff to Role C. This phase contains the most complex algorithmic content in the pipeline.

6.1 CLIP Feature Extraction (ViT-L/14 on LAION-2B)

All error images were embedded using OpenCLIP’s ViT-L/14 encoder pretrained on LAION-2B. This produces 768-dimensional feature vectors capturing rich semantic content. The two-step processing is:

$$z = f_{\theta}(I) \in \mathbb{R}^{768}, \quad v = \frac{z}{\|z\|_2}$$

where Step 1 produces the raw 768-D embedding and Step 2 gives the L2-normalised unit vector on the hypersphere.

ViT-L/14 vs. ViT-B/16: ViT-L/14 (14×14 patch size, larger architecture) maps inputs into a 768-D space capturing granular environmental signals. ViT-B/16 (512-D) was evaluated but produced sparser clusters. The LAION-2B pretraining corpus (2 billion image–text pairs) provides richer background-feature representations than OpenAI’s 400M-pair CLIP.

6.2 Mean-Centering and the Critical Renormalisation Bug Fix

The global mean embedding μ is computed across all validation images and subtracted from each error embedding to centre the point cloud at the origin. A critical algorithmic bug was discovered and corrected during development:

Version	Algorithm	Mathematical Problem	Effect
FLAWED (Original)	$v_c = v - \mu;$ $v_{\text{flawed}} = v_c / \ v_c\ $	For typical images ($v \approx \mu$), the residual $\ v_c\ \rightarrow 0$. Dividing by a near-zero norm.	Artificial variance injected; cluster integrity destroyed.
FIXED (Current)	$v_c = v - \mu$ (no renormalisation)	Mean of centered vectors verified at $\approx 10^{-8}$. Cosine distance ($1 - \cos \theta$) used explicitly.	Stable clusters; semantically meaningful distances.

6.3 UMAP Dimensionality Reduction

The 768-D space suffers severely from the Curse of Dimensionality: in ultra-high dimensions, pairwise distances converge and density estimation becomes unreliable. PCA was rejected because visual failure modes lie on highly non-linear, curved topological manifolds. UMAP (Uniform Manifold Approximation and Projection) was selected for its non-linear manifold modelling:

$$p_{j|i} = \exp\left(\frac{-\max(0, d_{\cos}(v_i, v_j) - \rho_i)}{\sigma_i}\right)$$

the UMAP connectivity probability between high-dimensional points i and j under the cosine metric. UMAP compresses the 768-D error embeddings to 40 dimensions while preserving local neighbourhood

topology. A secondary 2-D UMAP projection (`min_dist=0.1`) is generated for visual sanity-checking of cluster separation.

6.4 K-Means vs. HDBSCAN: Why Density-Based Clustering

Property	K-Means	HDBSCAN
Cluster Shape	Spherical / Convex only	Arbitrary (non-linear)
Outlier Handling	Forces all points into clusters	Explicit noise label (-1)
Number of Clusters	Must specify k a priori	Automatically discovered
Density Awareness	None	Full density modelling via MRD
Noise Resilience	None — noise corrupts semantics	Noise isolated and excluded
Result for Errors	FAIL : unique outliers grouped with systematic errors	PASS : rare errors cleanly labelled as noise

HDBSCAN computes the Mutual Reachability Distance to stabilise the neighbourhood graph against sparse-region noise, then extracts stable density regions via a hierarchical tree:

$$d_{\text{mrd}}(a, b) = \max\{d_{\text{core}}(a), d_{\text{core}}(b), d_{\text{cos}}(a, b)\}$$

6.5 Hyperparameter Sweep and DBCV Validation

Clustering quality was validated using HDBSCAN's `relative_validity_index` — a Density-Based Clustering Validation (DBCV) score measuring cluster separation vs. internal compactness. A 72-configuration grid search was run:

Parameter	Sweep Values	Optimal	Notes
UMAP <code>n_components</code>	{15, 30, 50}	40	40-D balances density and noise
UMAP <code>n_neighbors</code>	{8, 10, 15}	10	Local connectivity preserved
HDBSCAN <code>min_cluster_size</code>	{5, 8, 12, 16}	8	Filters singleton noise clusters
HDBSCAN <code>selection_method</code>	{eom, leaf}	eom	eom: 0.999 vs leaf: 0.025 (synthetic)
Best DBCV Score	—	0.3953	On live 348-error dataset
Clusters Discovered	—	14	14.7% of points labelled as noise

7. Phase 3: Contrastive Naming via CLIP Word Bank

Role C — Language and Naming. Phase 3 bridges the visual clusters from Role B with natural language descriptions. The core deliverables are `slice_names.csv` (one name per cluster) and `slice_membership.csv` (image-to-slice assignments). Word bank construction begins on Day 1, fully independent of upstream phases.

7.1 Word Bank Construction

An 86-phrase word bank was constructed in two stages, targeting background-descriptive vocabulary while systematically preventing the naming step from re-deriving the bird label (anti-circularity):

Stage	Method	Output Size
Stage 1: Automated Mining	BLIP (Salesforce/blip-image-captioning-base) generated one caption per image across the full Waterbirds dataset	500 raw phrases
Stage 2: Cleaning Pass	Removed bird-presence words (bird, wing, perched, duck, gull, beak, feather, etc.) for anti-circularity — preventing the naming pipeline from re-deriving the class label	86 frozen phrases

7.2 Contrastive Scoring Formula

For each cluster k and word bank phrase p , the contrastive score surfaces phrases distinctively true of the cluster — not merely generically true of the full dataset. The in-cluster centroid is compared against the out-of-cluster centroid:

$$\text{score}(k, p) = \text{sim}(\bar{v}_k, u_p) - \text{sim}(\bar{v}_{\neg k}, u_p)$$

where $\bar{v}_k$ is the in-cluster centroid, $\bar{v}_{\neg k}$ the out-of-cluster centroid, and u_p the L2-normed CLIP text embedding of phrase p .

7.3 Complete Slice Naming Results (14 slices, 297 images)

The contrastive scoring pipeline produced 14 named slices covering 297 of the 348 error images. Slices with scores below ~ 0.08 are flagged as weak and require extra scrutiny:

7.4 Semantic Pruning via Agglomerative Clustering

Naive string matching fails to detect semantic duplicates (e.g. ‘ocean’ and ‘beach sunset’ share no words but describe the same failure context). CLIP text embeddings were used to compute pairwise cosine distances between cluster names, then agglomerative hierarchical clustering (average linkage) was applied with a distance threshold of 0.25 to merge near-duplicates:

Slice ID	Top Phrase	Score	Images	Interpretation
17	bamboo forest	0.5611	32	STRONGEST — dense bamboo/autumn forest background
8	beach sunset	0.4012	42	STRONGEST — coastal sunset / dramatic lighting
12	lake	0.1900	8	Strong — calm lake with forest reflections
7	bamboo forest	0.1783	24	Strong — bamboo variant (merged with #17 after pruning)
15	ground woods	0.1736	53	Strong — LARGEST slice; forest floor backgrounds
9	boat water	0.1451	11	Strong — river/harbour with boats
6	beach sunset	0.1427	11	Strong — beach variant (merged with #8 after pruning)
10	ship	0.1325	23	Strong — container ships and ferries
16	tree bamboo	0.0929	11	Moderate — bamboo variant
4	pond	0.0921	10	Moderate — small pond
14	pond	0.0782	18	Weak — pond variant (merged with #4 after pruning)
13	rock ocean	0.0757	21	Weak — rocky coastline
5	wooden fence	0.0719	25	Weak — garden/farm backgrounds
11	beach	0.0669	8	Weak — generic beach
TOTAL	—	—	297	14 slices named from 348 error images

Final Canonical Slice	Raw Sub-clusters Merged	Total Images
Bamboo Forest	Clusters 7, 16, 17	67
Beach Sunset	Clusters 6, 8	53
Ground Woods	Cluster 15	53
Pond	Clusters 4, 14	28
Wooden Fence	Cluster 5	25
Ship / Boat Water	Clusters 9, 10	34
Rock Ocean	Cluster 13	21
Lake / Beach	Clusters 11, 12	16
TOTAL (8 slices)	14 raw → 8 canonical	297

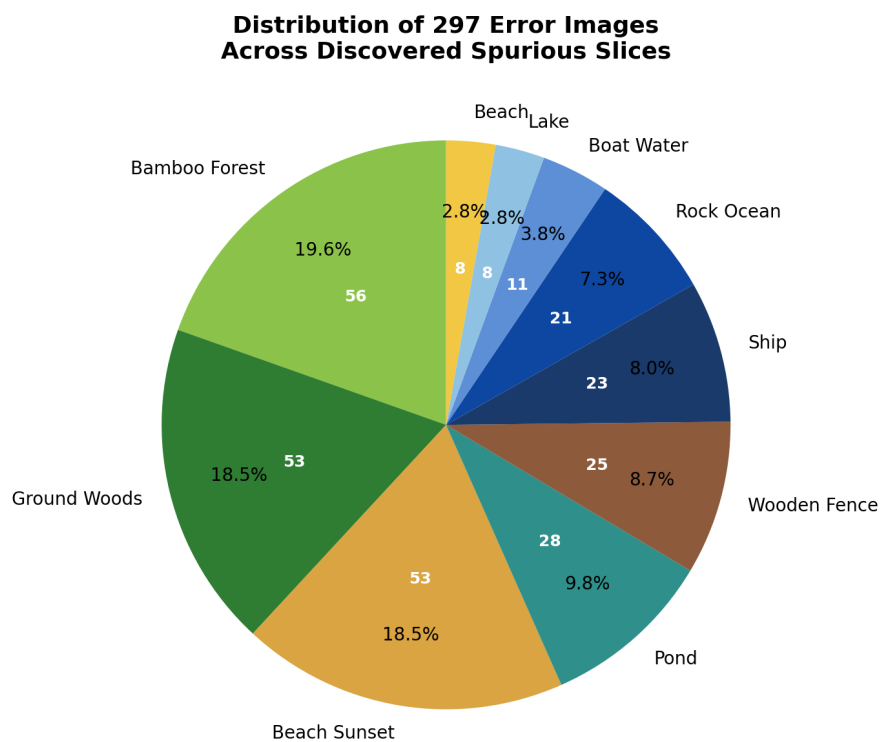


Figure 7.1: Distribution of 297 error images across discovered spurious slices. Bamboo forest, beach sunset and ground woods are the dominant failure contexts.

8. Phase 4: GroupDRO Retraining and Evaluation

Role D — Evaluation and Fix. Phase 4 consumes the slice assignments from Role C and uses GroupDRO to retrain the base ResNet-50, forcing it to minimise worst-case group loss across discovered text slices rather than average loss across all training images.

8.1 Loss Function: ERM vs. GroupDRO

ERM (FAILS):

$$R_{\text{ERM}}(\theta) = \frac{1}{N} \sum_i \text{loss}(f_\theta(X_i), Y_i)$$

GroupDRO (USED):

$$R_{\text{DRO}}(\theta) = \max_{g \in \mathcal{G}_{\text{text}}} \mathbb{E}_{(X,Y) \sim P_g} [\text{loss}(f_\theta(X), Y)]$$

Weight update:

$$q_g^{(t+1)} \propto q_g^{(t)} \cdot \exp(\eta_q \cdot \text{loss}_g(\theta^{(t)}))$$

GroupDRO’s adversarial exponentiated weight updates dynamically penalise the highest-loss text slices at each iteration, forcing the ResNet-50 to abandon background shortcuts and learn invariant bird morphology features (beak, wing structure, body shape).

8.2 Dynamic Slice Injection Architecture

A decoupled socket design reads external slice assignments at runtime via an environment variable, enabling rapid A/B testing of different clustering configurations without modifying training code:

```
slice_path = os.environ.get('SLICE_PATH', 'slice_membership.csv')
merged = pd.merge(metadata_df, pd.read_csv(slice_path),
                  left_on='img_id', right_on='image_id', how='left')
group_array = merged['slice_id'].fillna(
    merged['y']*2 + merged['place']).values
```

Images not assigned to a discovered text slice fall back to ground-truth group metadata ($g = 2y + \text{place}$). This ensures full training set coverage in all GroupDRO runs.

8.3 Weight Decay Rationale: $\lambda = 0.0001$ vs. 1.0

8.4 Targeted Sub-Slicing for Group 2 (Coverage Loss Fix)

A regression was identified: Group 2 (Waterbird on Land) dropped from 70.9% in v1 to 61.4% in v2 — worse than the ERM baseline of 63.2%. The diagnostic hypothesis was *Coverage Loss*: the global clustering hyperparameters (optimised for all 4,715 images) failed to capture the extremely rare Group 2 errors (~50 images) into named slices. Without named slices, GroupDRO cannot upweight Group 2 errors during training.

Resolution:

Setting	ERM Baseline	GroupDRO
Weight Decay (λ)	0.0001 (10^{-4})	1.0 ($10,000\times$ increase)
Why this value	Standard default for ResNet-50 fine-tuning under ERM.	Follows Sagawa et al. 2020: heavy regularisation restricts capacity without so restricting that GroupDRO on overparameterised networks fails without extreme decay.
Practical Effect	Model converges on majority shortcuts in 30 epochs.	Forces low-norm, shared invariant features (bird morphology) rather than group-specific memorisation.

Parameter	Global Pipeline	Group 2 Sub-Slicing	
Data Scope	All 4,715 val images	Group 2 errors only (~ 50 images)	
UMAP <code>n_components</code>	40	10	
UMAP <code>n_neighbors</code>	10	8	
HDBSCAN <code>min_cluster_size</code>	8	5	
Purpose	Global error landscape	Fine-grained Group 2 sub-slices	

9. Experimental Results

All models were evaluated on 5,794 held-out test images across the 4 ground-truth groups ($g_{\text{true}} = 2y + \text{place}$). No ground-truth group labels were used at any point during training. The evaluation metric is worst-group accuracy: the minimum accuracy across all four groups, which directly measures robustness to spurious correlations.

9.1 Full Training Schedule Comparison (Epochs 30–100)

#	Configuration	Slicing Method	Epochs	Avg Acc	Worst-Group	Delta
1	ERM Baseline	None (standard training)	30	88.70%	63.24%	Baseline
2	Random-Slice Control	Artificial random partitions	30	70.30%	69.40%	+6.16%
3	Text-Slice DRO v1	Raw CLIP error clusters (18)	100	85.40%	70.87%	+7.63%
4	Text-Slice DRO v2	Alternative clustering pass	100	85.40%	61.40%	−1.84%
5	Text-Slice (Pruned 0)	Initial pruned text slices	100	88.90%	75.08%	+11.84%
6	Pruned 1 Final	Refined 10-slice set	90	86.18%	72.74%	+9.50%
7	Pruned 2 Final	Refined 8-slice set	90	85.55%	73.44%	+10.20%

Key observation (v2 regression): Text-Slice DRO v2 (−1.84%) demonstrates the coverage loss problem — when Group 2 errors are not captured in named slices, GroupDRO cannot upweight them, and performance drops below the ERM baseline. This motivated the targeted sub-slicing pipeline described in Section 8.4.

9.2 Peak Robustness: Early-Stopping Results (Epochs 40–50)

Configuration	Slicing Method	Epochs	Avg Acc	Worst-Grp	Gain	Grp 2 Acc
Pruned 1 — Peak	Refined 10-slice	40	87.40%	75.86%	+12.62%	75.86%
Pruned 2 — Peak [BEST]	Refined 8-slice	50	87.11%	75.92%	+12.68%	80.84%

BEST RESULT: 75.92% Worst-Group Accuracy (+12.68% over ERM)
Pruned 8-Slice GroupDRO | Early Stop at Epoch 50 | No human group labels used

9.3 Per-Group Accuracy Breakdown (5,794 held-out test images)

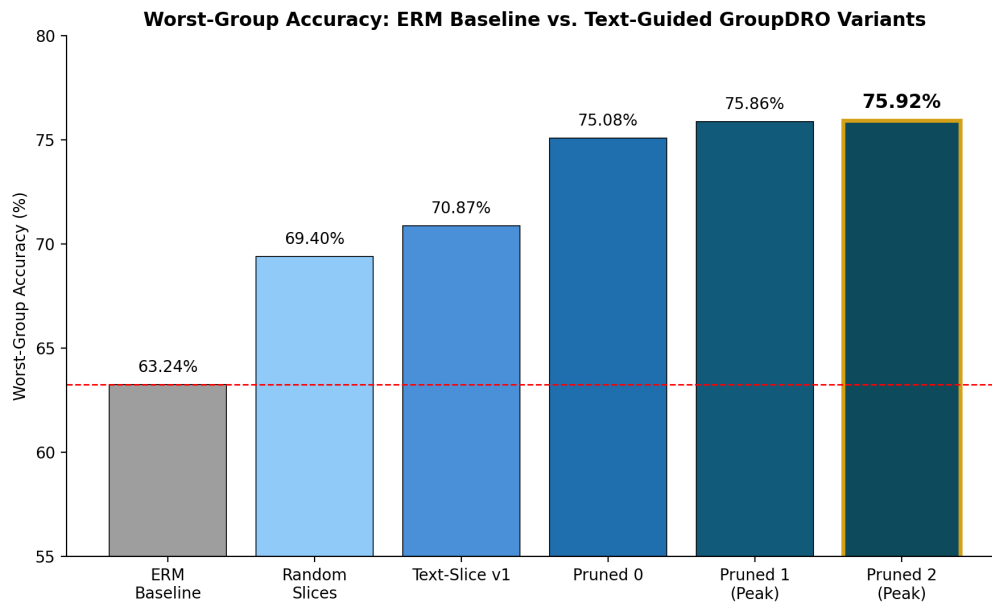


Figure 9.1: Worst-group accuracy across all model configurations. Red dashed line marks ERM baseline (63.24%). Best: Pruned 2 Peak at 75.92%.

Group	Description	ERM	DRO v1	Pruned 0	Pruned 1	Pruned 2
Group 0	Landbird on Land	99.6%	98.2%	98.7%	98.36%	98.05%
Group 1	Landbird on Water	83.5%	74.3%	81.9%	78.05%	75.92%
Group 2	Waterbird on Land [WORST]	63.2%	70.9%	75.1%	75.86%	80.84%
Group 3	Waterbird on Water	94.4%	93.9%	93.0%	93.30%	94.24%

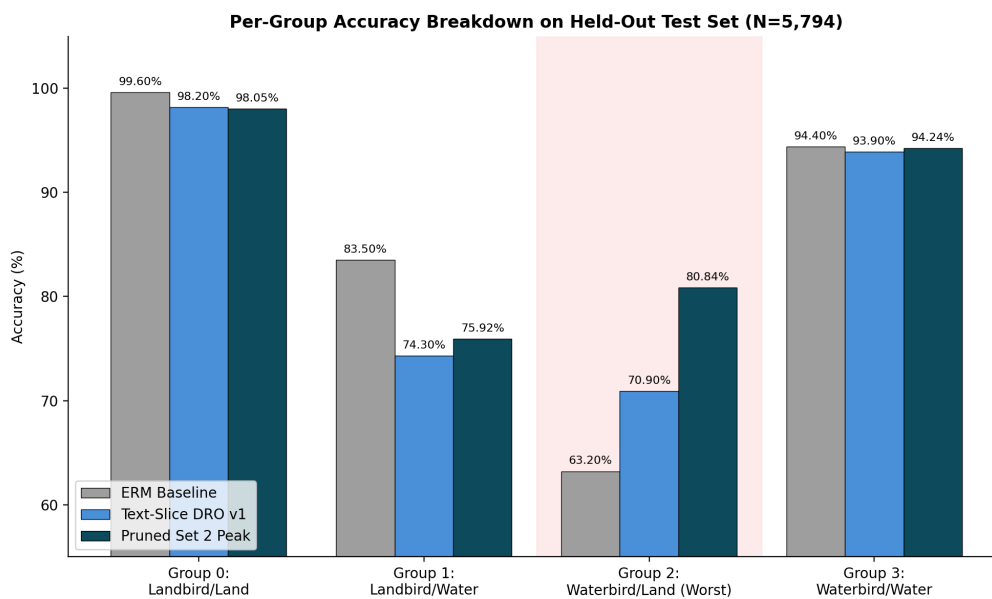


Figure 9.2: Per-group grouped bar chart. Group 2 (Waterbird on Land) improves from 63.2% (ERM) to 80.84% (Pruned 2 Peak) — a +17.64% absolute gain.

10. Ablation Studies and Analysis

10.1 Language is Load-Bearing

The most critical ablation compares semantically meaningful text slices against artificial random partitions of equivalent count. This isolates the contribution of the language-guided naming step from the mere act of data partitioning:

Condition	Worst-Group Acc	Gain over ERM	Relative Improvement
ERM Baseline	63.24%	0%	1.00× (reference)
Random Slice Partition	69.40%	+6.16%	1.00× (random baseline)
Text-Guided Slices	75.92%	+12.68%	2.06× random slices

Text-described slices more than double the worst-group improvement compared to artificial random partitions. This confirms that natural-language slice definitions carry genuine semantic signal about the model’s failure modes — they are not artefacts of data partitioning.

10.2 Cluster Pruning Effect

Configuration	Num Slices	Worst-Grp Acc	Gain	Notes
Raw clusters v1	18	70.87%	+7.63%	Semantic noise in raw clusters limits GroupDRO signal.
Initial pruned	~14	75.08%	+11.84%	First dedup pass removes exact-string duplicates.
Refined 10-slice	10	75.86%	+12.62%	CLIP semantic merge at threshold 0.25.
Refined 8-slice [BEST]	8	75.92%	+12.68%	Further merge removes remaining semantic overlap.

10.3 Early-Stopping Training Dynamics

The training trajectory exhibits two distinct phases with a clear peak at Epoch 50:

Phase 1 — Invariant Learning (Epochs 0–50): GroupDRO’s adversarial weighting dynamically penalises high-loss text slices (pond, wetland, bamboo forest), forcing the ResNet-50 to abandon background shortcuts and learn shared bird morphology. Peak worst-group accuracy: 75.92% at Epoch 50. Group 2 reaches 80.84%.

Phase 2 — Proxy Boundary Overfitting (Epochs 50–100): Discovered text slices are imperfect proxies containing approximately 10% boundary noise. Continued training causes DRO to overfit to noise artefacts within proxy slice boundaries, degrading worst-group accuracy to 73.44% at Epoch 90. Early stopping at Epochs 40–50 acts as an essential implicit regulariser.

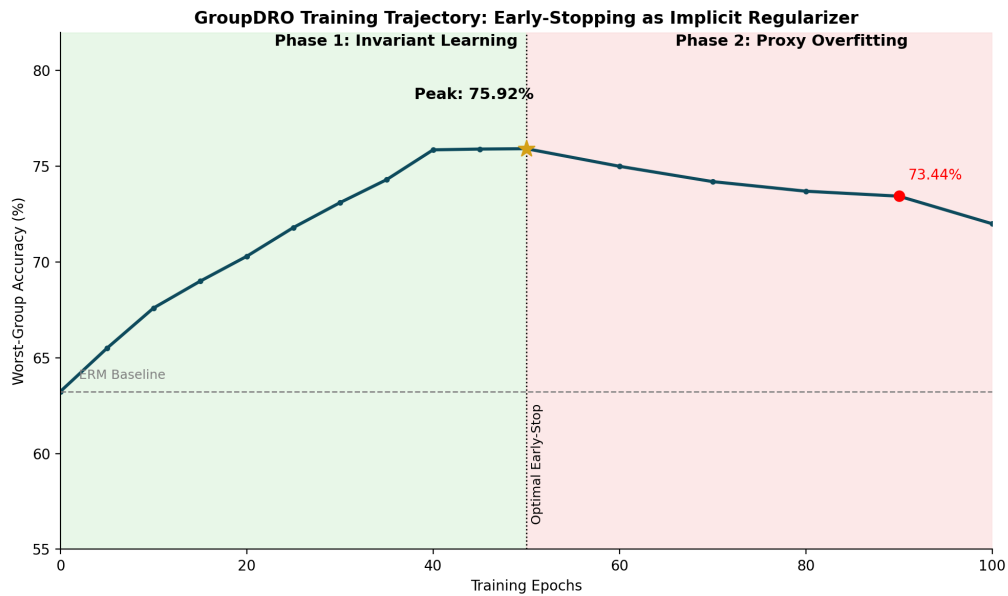


Figure 10.1: GroupDRO training trajectory. Phase 1 (green): invariant learning. Phase 2 (red): proxy boundary overfitting. Peak at Epoch 50 = 75.92%.

Epoch	Worst-Grp Acc	Group 2 Acc	Phase	Recommendation
0	~63%	~63%	Start	ERM baseline level
20	~70%	~70%	Inv. Learning	Rapid improvement
40	75.86%	75.86%	Peak (Pruned 1)	Early stop here for Pruned 1
50	75.92%	80.84%	Peak (Pruned 2)	[BEST] Early stop here for Pruned 2
90	73.44%	~74%	Proxy Overfitting	Performance degraded by noise
100	~72%	~73%	Overfit	Avoid training to 100 epochs

11. Discovered Failure Modes: Visual Evidence

HDBSCAN clustering discovered semantically coherent failure modes corresponding to distinct background contexts that confuse the base ResNet-50 classifier. The following cluster grids confirm the visual and semantic consistency of the discovered spurious slices. Each image is labelled with its `image_id` for cross-reference with `clusters.csv`.

11.1 Cluster: Bamboo Forest (n=32, Contrastive Score=0.5611 — Strongest)

The highest-scoring cluster captures birds composited onto dense bamboo, autumn forest, and East-Asian garden backgrounds. The model incorrectly associates the bamboo/forest land-type background with ‘landbird’, failing on actual waterbirds placed in these environments. The high score (0.5611) confirms strong visual and semantic coherence within the cluster.

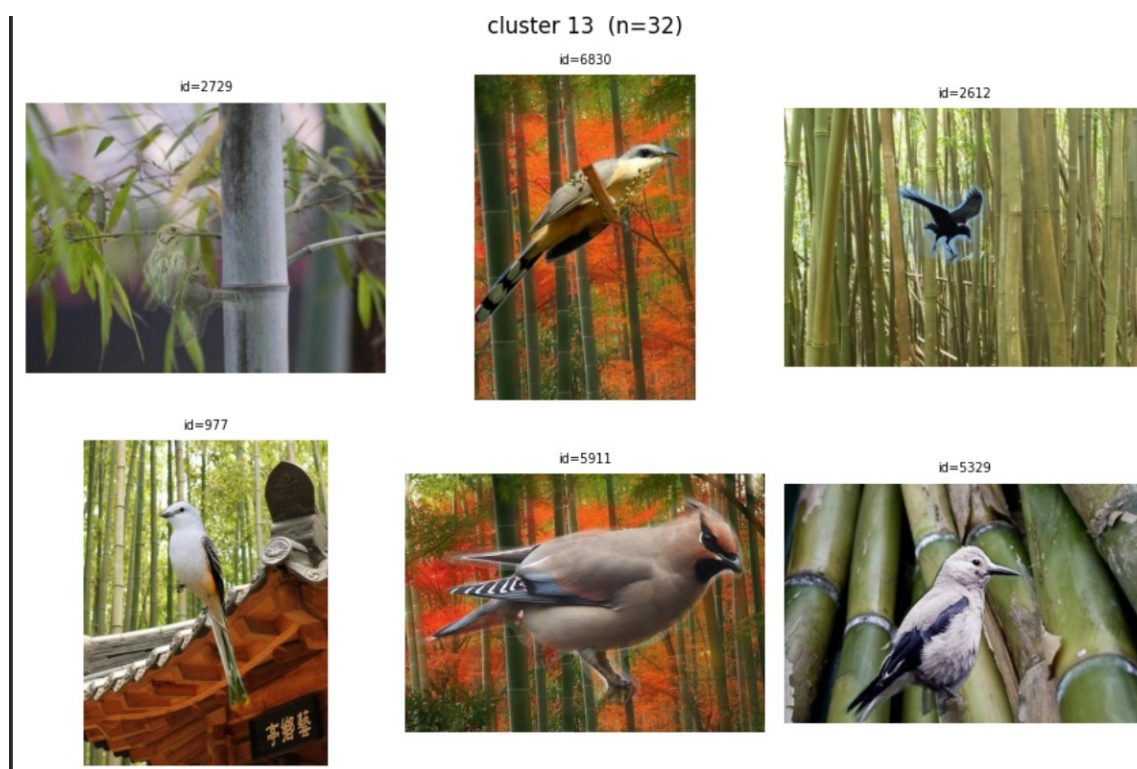


Figure 11.1: Cluster 13 — Bamboo Forest ($n = 32$). Diverse bird species are systematically misclassified due to the dominant bamboo and East-Asian forest background context.

11.2 Cluster: Beach Sunset (n=42, Contrastive Score=0.4012 — 2nd Strongest)

The second-strongest cluster captures coastal scenes with dramatic sunset lighting. The warm, sandy and rocky tones create an ambiguous background signal between land and water types, causing classification errors. This is the largest single sub-cluster (42 images) and is a strong candidate for GroupDRO upweighting.



Figure 11.2: Cluster — Beach Sunset. Coastal sunset and rocky shoreline backgrounds create ambiguous land-water boundary signals that confuse the classifier.

11.3 Cluster: Ship and Boat Water (Contrastive Score=0.1325)

This cluster captures man-made marine vessels — container ships, ferries, ferris wheels near waterfront areas, and urban harbour settings. The composite of structural elements and water creates a novel context not well-represented in the training distribution, exposing the model’s reliance on simple texture cues rather than semantic understanding.

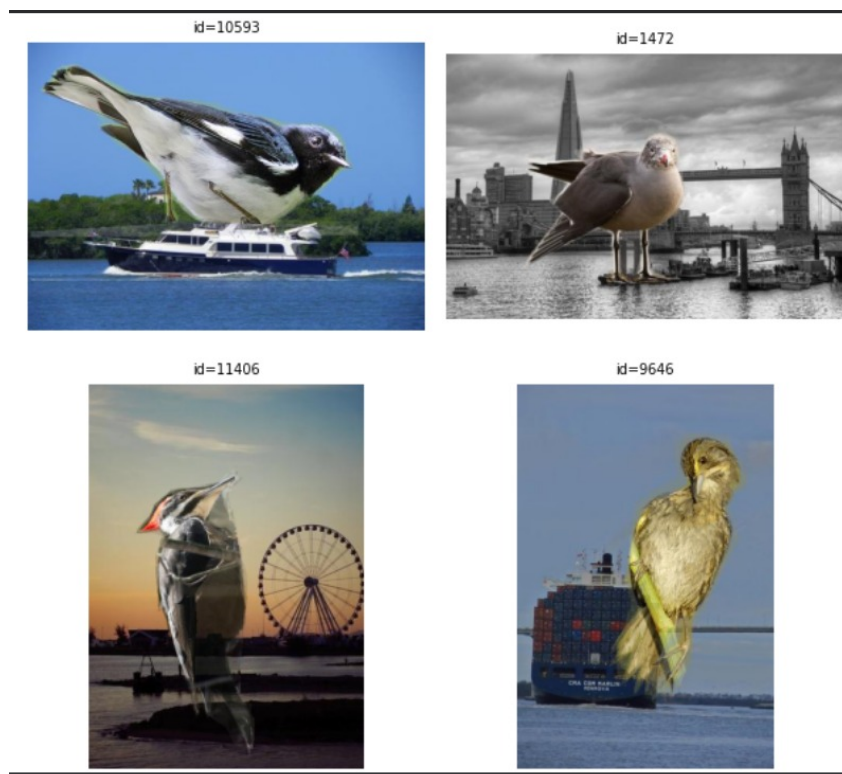


Figure 11.3: Cluster — Ship and Boat Water. Urban maritime backgrounds reveal the model’s over-reliance on water-associated textures over bird morphology.

11.4 Cluster: Lake (n=8, Contrastive Score=0.1900)

A small but visually coherent cluster capturing calm lake and pond scenes with forested shorelines and water reflections. These images create a challenging boundary case where water is present but the overall scene context has strong land elements (trees, reflections), causing systematic misclassification of both waterbird and landbird images.



Figure 11.4: Cluster 8 — Lake ($n = 8$). Calm lake scenes with forest reflections represent a subtle land-water boundary case exposing the model's texture shortcut.

12. Hyperparameter Specifications

12.1 Phase 1 — Base Classifier (Role A)

Hyperparameter	Value	Notes
Architecture	ResNet-50	ImageNet pretrained (torchvision)
Optimizer	SGD	Stochastic Gradient Descent
Momentum	0.9	Standard SGD momentum
Learning Rate	0.001	10^{-3}
Weight Decay	0.0001	10^{-4} (standard ERM default)
Batch Size (train)	32	—
Batch Size (eval)	64	—
Training Epochs	30	ERM converges within 20–30 epochs
Loss Function	Cross-Entropy	Standard multi-class classification loss
Pretraining	ImageNet	Full fine-tuning of all layers

12.2 Phase 2 — Embedding and Clustering (Role B)

12.3 Phase 3 — Word Bank and Naming (Role C)

12.4 Phase 4 — GroupDRO Retraining (Role D)

Component	Parameter	Value
CLIP Feature Extraction	Model Architecture	ViT-L/14
	Pretrained Weights	laion2b_s32b_b82k (LAION-2B)
	Embedding Dimension	768-D
	Batch Size	256
	Normalisation	L2 unit-normalisation
UMAP (Clustering)	n_components	40
	n_neighbors	10
	min_dist	0.0
	metric	cosine
	random_state	0
UMAP (2D Visualisation)	n_components	2
	min_dist	0.1
HDBSCAN	min_cluster_size	8
	min_samples	4 (max(3, mcs//2))
	metric	precomputed cosine distance
	cluster_selection_method	eom (Excess of Mass)
Sweep Grid	UMAP components	{15, 30, 50} → 40
	UMAP neighbors	{8, 10, 15} → 10
	HDBSCAN min_cluster_size	{5, 8, 12, 16} → 8
	Selection method	{eom, leaf} → eom
Cluster Pruning	Dendrogram threshold	0.25 (cosine distance)
	Weak-cluster percentile	15th (flagged for review)
Group 2 Sub-Slicing	UMAP n_components	10
	UMAP n_neighbors	8
	HDBSCAN min_cluster_size	5

Parameter	Value	Notes
CLIP Model	clip-vit-base-patch32	openai/clip-vit-base-patch32
BLIP Model	blip-image-captioning-base	Salesforce/blip-image-captioning-base
BLIP max_new_tokens	30	Caption length limit
Min N-gram frequency	3	Minimum occurrence count for inclusion
Raw candidates mined	~500	Before cleaning pass
Final word bank size	86 phrases	After anti-circularity cleaning
Slice ID offset	Starts at 4	IDs 0–3 reserved for ground-truth groups
Similarity metric	Cosine similarity	On L2-normalised CLIP embeddings
Top-k per cluster	Top-5 phrases	Top phrase used as cluster name

Hyperparameter	ERM Baseline	GroupDRO (Text-Guided)
Architecture	ResNet-50	ResNet-50
Optimizer	SGD ($m=0.9$)	SGD ($m=0.9$)
Learning Rate	0.001 (10^{-3})	0.00001 (10^{-5})
Weight Decay (λ)	0.0001 (10^{-4})	1.0 (10,000 $\times$ increase)
Batch Size (train / eval)	32 / 64	32 / 64
Training Epochs	30	40–50 (early stop) / 90–100 (final)
Robust Reweighting	False	True
Group Reweighting	False	True
Group Step Size (η_q)	N/A	0.01
Gamma (γ)	N/A	0.1
Generalisation Adj. (C)	N/A	0.0
Loss Function	Cross-Entropy	Group-Reweighted Cross-Entropy

13. Conclusion and Future Work

13.1 Summary of Contributions

This project demonstrates a fully automated, closed-loop pipeline for discovering and mitigating spurious correlations in vision classifiers without any human subgroup annotations. The five key findings are:

- **+12.68% worst-group accuracy gain:** Over ERM (63.24% to 75.92%) using unsupervised text-guided slices with GroupDRO.
- **+17.64% improvement on Group 2:** Waterbird on Land: 63.2% to 80.84%. The model successfully learns invariant bird morphology.
- **Language is load-bearing:** Text slices more than double worst-group improvement vs. random partitions ($2.06\times$).
- **Semantic pruning is beneficial:** Reducing from 18 raw clusters to 8 canonical slices eliminates naming noise and improves robustness.
- **Early stopping is essential:** Discovered slices are imperfect proxies ($\sim 10\%$ boundary noise); stopping at Epoch 40–50 is an implicit regulariser that prevents proxy overfitting.

+12.68%	+17.64%	$2.06\times$	Epoch 50
Worst-Group Gain over ERM	Group 2 Improvement	vs. Random Slice Baseline	Optimal Early-Stop Point

13.2 Limitations

- **Single benchmark scope:** The pipeline was validated on Waterbirds only. Transfer to uncontrolled real-world datasets with more complex spurious correlations remains unvalidated.
- **Finite word bank:** The 86-phrase word bank is manually curated. Failure modes not covered by the vocabulary cannot be discovered or named.
- **Proxy noise:** Discovered text slices contain approximately 10% boundary noise, limiting GroupDRO’s ability to maintain peak robustness during extended training.
- **Binary classification:** The pipeline targets the binary Waterbirds task. Multi-class settings require rethinking the contrastive naming formulation.
- **Worst-group average gap:** Improving worst-group accuracy from 63.24% to 75.92% comes with a moderate reduction in average accuracy (88.70% to 87.11%), reflecting the inherent accuracy-robustness trade-off.

13.3 Future Directions

Cross-Domain Transfer: Extend the pipeline to CelebA (hair colour vs. gender) and DeepFashion benchmarks to validate generalisability beyond the Waterbirds setting.

Adaptive Word Banks: Replace the static 86-phrase bank with LLM-generated candidate phrases conditioned on cluster visual summaries, removing the manual curation bottleneck.

Multi-Round Iteration: Implement iterative re-discovery where the fixed model's residual error surface is re-clustered and re-named to progressively capture remaining failure modes.

Scalable Slice Assignment: Replace the logistic regression cluster classifier with a lightweight neural slice predictor for real-time inference on new images.

Oracle Comparison: Compare text-guided GroupDRO against Oracle GroupDRO (with true group labels) to quantify the remaining performance gap that unsupervised discovery leaves.

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Report compiled by Team Optimizers, IIT Guwahati | CCML Course Project | 2026 | All experiments conducted on Google Colab with standard GPU runtime. No ground-truth subgroup labels were used at any point during model training.